Block 4 – Session 1

Data Centers & Network Virtualization

Arquitectura de Xarxes

Universitat Pompeu Fabra

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Outline

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What Is the Cloud and How Is It Formed?

Every time you use Netflix, Gmail, or ChatGPT,
your request travels to a building full of servers.
What happens inside that building?

Learning objectives:

1. Understand why data centers exist and what they contain
2. Explain virtualization and distinguish VMs, hypervisors, and containers
3. Describe how networks are virtualized (vNICs, vSwitches, overlays)

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Block 4 Roadmap

	Session 1 (today)	**Session 2**
Theme	*The infrastructure*	*The business model*
Topics	Data centers VMs & hypervisors VMs vs containers (overview) vNICs, vSwitches, vRouters Multi-tenancy & VLANs	Overlay networks (VXLAN) From virtualization to cloud NIST cloud definition IaaS / PaaS / SaaS VPC architecture Security groups & ACLs Governance & GDPR

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What Is a Data Center?

- A **data center** is a facility housing computer systems and networking equipment
- Purpose: run applications and store data **reliably, at scale, 24/7**
- Ranges from small server rooms to **warehouse-scale** buildings

Who operates them?

- **Enterprises**: banks, hospitals, universities (their own IT)
- **Cloud providers**: AWS, Azure, GCP (sell capacity to others)
- **Colocation**: you rent space; provider handles power and cooling

Source: Barroso, Clidaras & Hölzle, *The Datacenter as a Computer*, 3rd ed., Morgan & Claypool, 2019.

A Data Center from the Inside

Rows of server racks, each containing dozens of servers, connected by structured cabling. Source: Cisco, *What Is a Data Center?*, cisco.com.

Inside a Data Center: Five Pillars

- **Compute**: CPUs that execute workloads (web servers, databases, analytics)
- **Accelerators**: GPUs and TPUs for AI/ML training and inference
  - The explosive growth of AI has made GPUs a critical data center resource
- **Storage**: SSDs and HDDs holding persistent data
- **Network**: switches, routers, and structured cabling connecting everything
- **Facilities**: redundant power supplies, cooling systems, physical security

The Problem with Physical Infrastructure

- Traditional model (1990s–early 2000s): **one server = one application**
- Typical server utilization: only **10–15%** of capacity
- Adding capacity means buying, racking, cabling → **weeks or months**
- Hardware failure takes down the entire service
- No way to scale down when demand drops

Key insight: most servers sit idle most of the time. We are paying for hardware we barely use.

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The Cloud: From Waste to Efficiency

- **1990s–early 2000s**: one application per server; most capacity wasted
- **2001**: VMware makes it possible to run **multiple systems on one machine**
  - Technology: **Virtual Machines (VMs)**
- **Mid-2000s**: enterprises consolidate servers (60–80% utilization)
- **2006–2010**: Amazon, Google, and Microsoft start **renting computing capacity** online
  - Technology: **Cloud Computing** (AWS, Azure, GCP)
- **2010s–today**: lightweight alternatives emerge; automation at massive scale
  - Technologies: **Containers** (Docker) and **orchestration** (Kubernetes)

We will explore each of these technologies step by step throughout Blocks 4 and 5.

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Scalability and Elasticity

- **Scalability**: ability to grow resources as demand increases
- **Elasticity**: ability to grow *and shrink* automatically
- Physical infrastructure provides **neither** in practice:
  - Cannot add servers in minutes
  - Cannot return servers when demand drops
- Virtualization enables both $\rightarrow$ foundation of **cloud computing** (Session 2)

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Discussion: Data Centers

A company runs 50 servers, each at 10 percent utilization.
That is 50 machines doing the work of 5.
What would you do to reduce waste?

Hints: consolidation, fewer physical machines, higher utilization, isolation between workloads.

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What Is Virtualization?

- **Virtualization** makes it possible for a single computer to act like many
- A software layer called **hypervisor** divides the physical resources (CPU, RAM, disk, NIC) into isolated **virtual machines (VMs)**
- Each VM gets its own share and believes it owns dedicated hardware

Why virtualize?

- **Resource efficiency**: consolidate workloads instead of one app per server
- **Cost savings**: fewer servers, less power, cooling, and space
- **Flexibility**: create or destroy environments in minutes
- **Isolation**: VMs are independent; failures do not propagate

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Virtual Machines (VMs)

- A VM is an **isolated computing environment** with its own CPU, memory, network interface, and storage
- Runs a **full operating system** (Linux, Windows, etc.)
- Completely **isolated** from other VMs on the same host

Key properties:

- **Encapsulation**: entire VM state stored as files → easy to copy, move, backup
- **Hardware independence**: the VM does not care about the underlying hardware
- **Snapshotting**: save and restore VM state at any point in time

Source: Popek & Goldberg, "Formal Requirements for Virtualizable Third Generation Architectures," *CACM*, 1974.

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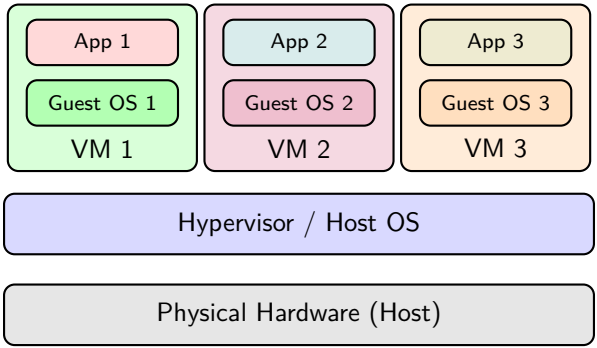
Host and Guest Systems



- **Host**: physical machine providing CPU, RAM, disk, NIC
- **Guest**: virtual machine running its own OS on the host
- Each guest believes it has **dedicated hardware** (abstraction)

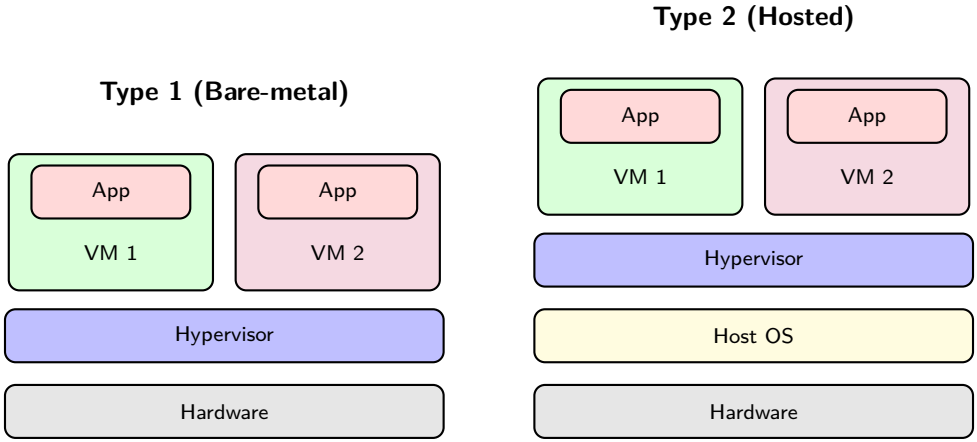
Hypervisors: The Key Enabler

- A **hypervisor** (or VMM) manages and allocates physical resources to VMs
- Two types:

	Type 1 (Bare-metal)	**Type 2 (Hosted)**
Runs on	Directly on hardware	On top of a host OS
Performance	Near-native	Some overhead
Use case	Data centers, production	Development, testing
Examples	VMware ESXi, KVM, Hyper-V	VirtualBox, VMware Workstation

Source: Barham et al., "Xen and the Art of Virtualization," *SOSP*, 2003.

Type 1 vs Type 2: Visual Comparison



- Type 1: hypervisor runs **directly on hardware** (no host OS) → less overhead, used in production
- Type 2: hypervisor runs **on top of a host OS** → easier to set up, used for dev/testing

VMs vs Containers: Overview

Virtual Machines

- Include a **full guest OS** per instance
  - A typical OS: 1–5 GB just for the OS itself
- Size: **GBs** (OS + app + dependencies)
- Boot time: **minutes**
- Strong isolation (hardware level)
- Can run **different operating systems**

Containers

- **No guest OS**; share the host kernel
- Size: **MBs** (app + dependencies only)
- Boot time: **seconds**
- Good isolation (kernel namespaces/cgroups)
- All containers share the **same OS kernel**

This is just an overview. We will do a deep dive into container internals and networking in **Block 5**.

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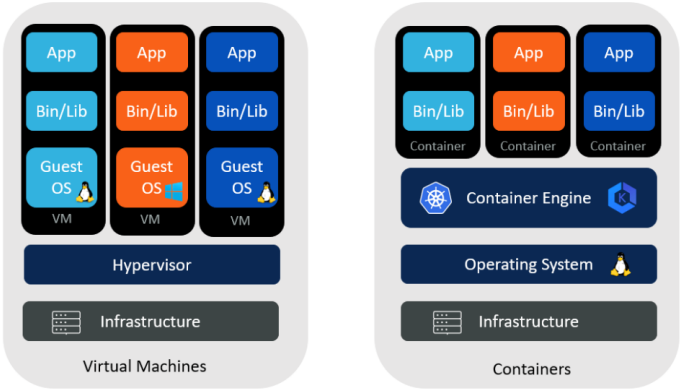
VMs vs Containers: Visual Comparison



Bins/Libs = binaries and libraries: the dependencies each app needs to run (e.g., Python, OpenSSL, libc).

Source: Aqueduct Technologies, "Containers and Virtual Machines," aqueducttech.com.

When to Use VMs vs Containers

- **Use VMs when** the app requires it:
  - Needs a **different OS** than the host (e.g., Windows app on a Linux host)
  - Requires **strong tenant isolation** (e.g., multi-tenant cloud)
  - **Legacy software** tied to a specific OS version
- **Use containers when** the app allows it:
  - Cloud-native or **microservices** applications
  - Need for **fast scaling** (seconds, not minutes)
  - **CI/CD pipelines**: build, test, deploy in identical environments
- **The trend**: most new applications are designed for containers
  - VMs remain essential for legacy workloads and strong isolation
  - Common pattern: containers run **inside VMs** (best of both worlds)

Microservices and container networking: **Block 5**.

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Discussion: Virtualization

You need to deploy 200 instances of the same web application that must scale up and down every few minutes. Would you use VMs, containers, or both? Why?

Hints: boot time, resource overhead, isolation needs, how fast you need to scale.

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From Physical to Virtual Networks

- You already know how physical networks work (Blocks 1 to 3):
  - NICs, switches, routers, VLANs, routing protocols
- When we virtualize servers, we also need to **virtualize the network**
- The same concepts apply, but implemented **in software** instead of hardware

Physical	**Virtual equivalent**
NIC (network interface card)	**vNIC** (virtual NIC)
Switch	**vSwitch** (virtual switch)
Router	**vRouter** (virtual router)
Cable	Internal software path

Key idea: if VMs think they have real hardware, they also need a **real-looking network**.

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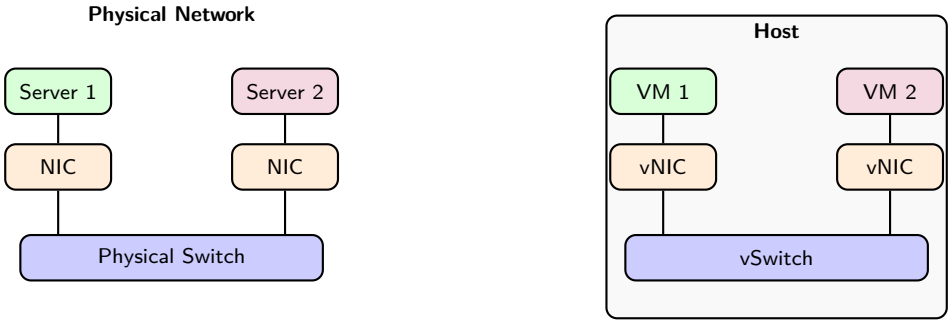
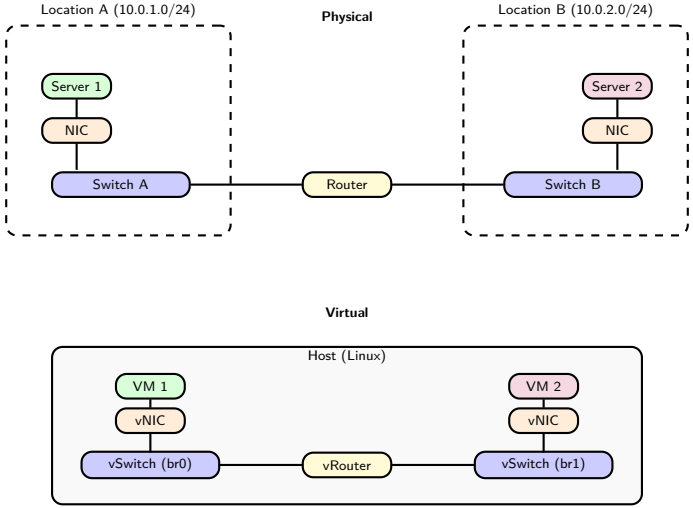
Physical vs Virtual: Side by Side (Same Network)



- Left: two physical servers connected by a physical switch and cables
- Right: two VMs connected by a virtual switch, all inside **one physical host**

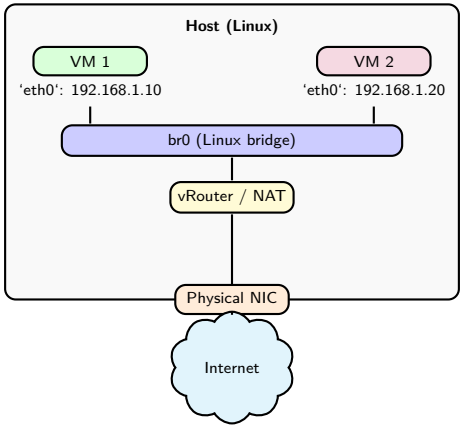
Physical vs Virtual: Side by Side (Different Networks)



Different subnets → router needed (L3). Physical uses hardware; virtual uses software inside one host.

Example: VM-to-VM Communication

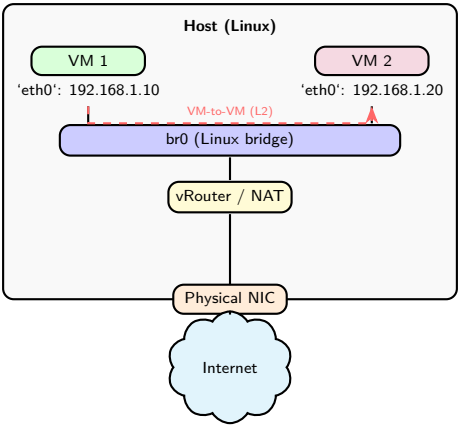
- Linux can act as a **Layer 2 switch** using a **bridge** (`br0`)
- VMs on the same bridge share a subnet: frames forwarded by MAC address



How does VM 1 reach VM 2? And how does it reach the Internet?

Example: VM-to-VM Communication (Solution)

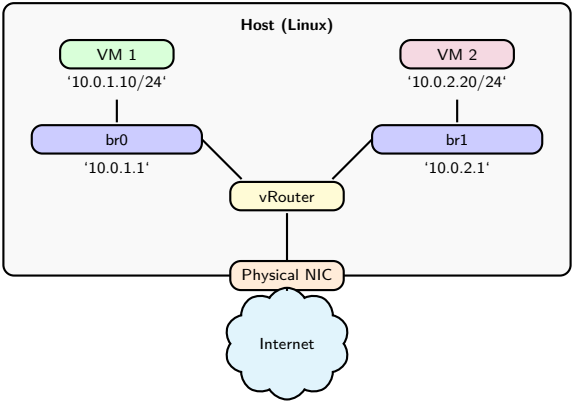
- Same subnet, same bridge → **L2 forwarding** (no routing needed)



Both VMs share `br0`: the bridge forwards frames by MAC address, just like a physical switch. No routing needed.

Example: VMs on Different Subnets

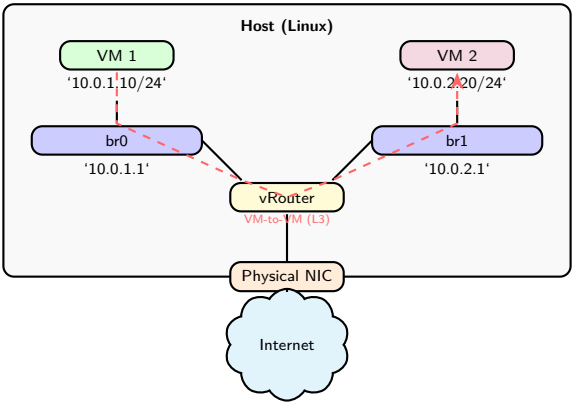
- Now VMs are on **different subnets**: they need L3 routing
- Two bridges (`br0`, `br1`), each with its own subnet



How does VM 1 (10.0.1.10) reach VM 2 (10.0.2.20)? What components are involved?

Example: VMs on Different Subnets (Solution)

- Different subnets → traffic must go through the **vRouter** (L3 forwarding)



Each VM has a default gateway pointing to its bridge IP. The host routes packets between subnets, just like a physical router.

Virtual Network Interfaces (vNICs)

- A **vNIC** is a software-emulated network interface card
- From the guest's perspective: behaves exactly like a real NIC
- Each VM gets **one or more vNICs**

Key characteristics:

- Has its own **MAC address** (software-assigned by the hypervisor)
- Has its own **IP address** configuration
- Connected to a **virtual switch** on the host
- Multiple vNICs per VM enable **multi-network** connectivity

Same MAC/IP concepts from Blocks 1 to 3, now virtualized inside the hypervisor.

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Types of Virtual Switches

Type	Scope	Example
Standard vSwitch	Single host	VMware vSS, Linux bridge
Distributed vSwitch	Multiple hosts, central mgmt	VMware vDS, Hyper-V
SDN-based vSwitch	Programmable, software-defined	OVS, VMware NSX

- All types support **VLANs** (same 802.1Q from Block 3, now inside the hypervisor)
- OVS is open source and **programmable via software** → deep dive in **Block 5** (SDN)

Source: Pfaff et al., "The Design and Implementation of Open vSwitch," *NSDI*, 2015.

Bridging Modes

Three ways to connect a VM to the outside world:

Mode	VM visibility	External access?	Use case
Bridged	Same subnet as host	Yes (direct)	Production servers
NAT	Hidden behind host IP	Outbound only	Dev/testing
Host-only	Only sees the host	No	Isolated labs

- **Bridged**: VM appears directly on the physical network
- **NAT**: VM traffic translated through the host's IP (same NAT from Block 3)
- **Host-only**: completely isolated from external networks

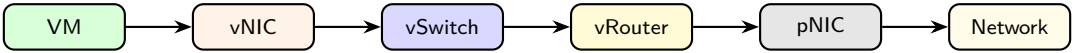
Virtual Routing and NAT

- A **virtual router** forwards traffic between virtual subnets
- Same function as a physical router, implemented **in software**
- Handles:
  - Routing between VMs on **different subnets**
  - **NAT** for VMs accessing external networks
  - Acting as **default gateway** for VMs

Same NAT concept from Block 3, now managed by the hypervisor instead of a physical device.

In Session 2 we will see how cloud providers offer NAT as a **managed service** (NAT Gateway).

Packet Flow: Virtual to Physical



1. VM generates packet with **virtual MAC/IP** as source
2. vNIC passes frame to **vSwitch** (L2 forwarding)
3. If destination is another subnet → **vRouter** (L3 forwarding)
4. vRouter may apply **NAT** (replace private IP with host IP)
5. Frame exits through **physical NIC** to the external network

Discussion: Virtual Networking

Two VMs on the same host want to communicate.
Does their traffic ever leave the physical machine?
What if they are on different subnets?

Hints: vSwitch handles same-subnet traffic locally; vRouter needed for different subnets; traffic may still stay inside the host.

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Putting It All Together

- We have seen the **building blocks**: vSwitches, vRouters, vNICs, VLANs
- These components do not run in isolation; they are managed by **platforms**
- Platforms orchestrate VMs or containers and configure virtual networking under the hood

Two categories:

- **VM management platforms**: create and manage virtual machines at scale
- **Container orchestration platforms**: deploy and scale containerized applications

Cloud Management Platforms (VMs)

- VMs need a **management layer** to orchestrate resources
- These platforms let you build and manage your **own (private) cloud**

Platform	Description	Logo
OpenStack	Open-source, widely used in private clouds	openstack.
VMware vSphere	Enterprise leader, proprietary	vmware
Proxmox VE	Open-source, lightweight alternative	PROXMOX

As an alternative: **public cloud providers** (pay-per-use, no hardware to manage)

Provider	Logo
Amazon Web Services (AWS)	aws
Microsoft Azure	Azure
Google Cloud Platform (GCP)	Google Cloud

Container Orchestration Platforms

- Containers need **orchestration** to manage hundreds or thousands of instances
- Key tasks: scheduling, scaling, networking, self-healing

Platform	Description	Logo
Docker	Standard container runtime; package and run containers	
Kubernetes (K8s)	De facto standard for orchestration at scale	
OpenShift	Enterprise K8s distribution by Red Hat	

Discussion: Platforms

A company runs 50 legacy Windows applications and 200 microservices. Which platform(s) would you recommend and why?

Hints: VMs for legacy Windows apps (different OS); containers for microservices (fast scaling); possibly both on the same infrastructure.

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Why Isolation Matters

- Data centers host workloads from **multiple tenants** (users, departments, companies)
- **Multi-tenancy**: shared infrastructure, but each tenant expects:
  - **Performance isolation**: my traffic is not affected by yours
  - **Security isolation**: you cannot see or access my data
  - **Address independence**: we can both use `10.0.0.0/24`

Without isolation: broadcast storms, ARP spoofing, IP conflicts, resource starvation.

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VLANs in Virtual Environments

- Same **802.1Q** concept from Block 3, now applied to virtual switches
- Hypervisors assign VMs to specific VLANs via **vSwitch port groups**
- Traffic tagged with VLAN IDs: tenants separated at L2

Limitation: VLAN ID is 12 bits $\rightarrow$ max **4,096 VLANs**

Problem: 4,096 VLANs is **not enough** for a large cloud with thousands of tenants.

Source: IEEE 802.1Q-2022, "Bridges and Bridged Networks."

Micro-Segmentation

- Beyond VLANs: **per-VM or per-workload firewall policies**
- Goal: **least privilege**, each VM should only reach what it strictly needs
- Key concept in modern **zero trust** security architectures

In Session 2 we will see how cloud providers implement this with **Security Groups** (per-instance) and **Network ACLs** (per-subnet).

Discussion: Network Isolation

You manage a data center with 5,000 tenants.
Each tenant wants their own isolated network.
Can you use VLANs alone? Why or why not?

Hints: think about the VLAN ID limit (4,096), overlapping IP ranges, and what happens when VMs move between hosts.

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Key Takeaways

1. **Data centers** house compute, accelerators (GPUs), storage, network, and facilities
2. Physical infrastructure is **rigid, underutilized, and slow to scale**
3. **Hypervisors** (Type 1 and 2) enable multiple VMs on one host
4. **Containers** are lighter than VMs but share the kernel (deep dive in Block 5)
5. Virtual networks mirror physical ones: **vNICs, vSwitches, vRouters**
6. **Multi-tenancy** requires isolation $\rightarrow$ VLANs and micro-segmentation

Next Session: Preview

In **Session 2** we build on today's foundation:

- **Overlay networks** (VXLAN): scaling beyond the 4,096 VLAN limit
- Virtualization is the technology; cloud is the **business model**
- NIST definition and **5 essential characteristics**
- Service models: **IaaS / PaaS / SaaS**
- **VPC** architecture: subnets, route tables, gateways
- Cloud security: **Security Groups + Network ACLs**
- Governance, **data sovereignty**, GDPR

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Discussion

A company runs 200 VMs on 10 physical servers.
Each team wants an isolated network.
What virtual networking components do you need
and how would you connect them?

Hints: think about vNICs, vSwitches, VLANs, and routing between subnets.

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